CHRIST (Deemed to be University)

Department of Computer Science & Engineering

■ EduBase – Full-Stack Student Management Portal

Software Engineering Process Model Implementation & Technical Report

Team Member	Engineering Role	Core Responsibilities
Yashwanth	Lead System Architect	Express 5 REST APIs, Dual JWT Security, MongoDB Relational Models, Caching Engine & SSE
Gagan Aditya	UI/UX & Frontend Lead	Design System, WebGL Shaders, React Architecture, Framer Motion Physics & Component Design

1. Executive Summary

EduBase is an enterprise-grade full-stack student management application engineered using **Node.js, Express 5, MongoDB Atlas, React 18, TypeScript, Tailwind CSS, Framer Motion, and Docker**. The application was designed to meet strict Software Engineering (SE) quality standards, employing the **Agile Model (Scrum Framework)** for sprint-based project delivery, paired with **Component-Based Development (CBD)** for modular UI and backend architecture.

2. Software Process Model Mapping & Defense

2.1 Primary Model: Agile Model (Scrum Framework)

Category in Syllabus: *4. Dynamic Models: Agile Model (Scrum)*

- **Adaptability & Iterative Slices:** Built over 3 dedicated sprints delivering working software increments at the end of each cycle.
- **Role Mapping:** Development team operated as a two-member Scrum feature team maintaining clear product backlog items.
- **Definition of Done (DoD):** Verified through 100% automated integration test suite execution and clean Docker container builds.

2.2 Secondary Model: Component-Based Development (CBD)

Category in Syllabus: *3. Specialized Process Models: Component Based Development*

- **Frontend Modularity:** Independent React components (StudentForm, StudentList, DepartmentChart, AuthPage, WebGL Shaders).
- **Backend Micro-Services Layering:** Decoupled route handlers, controllers, Mongoose schemas, and security middleware.

2.3 Prescriptive Fallback: Incremental Process Model

Category in Syllabus: *1. Prescriptive Process Models: Incremental Model*

- **Increment 1:** Baseline REST API (/api/v1) and core database schemas.

- **Increment 2:** Dual JWT Auth, Social OAuth 2.0, Glassmorphism UI, and Soft Delete Trash Console.
- **Increment 3:** API v2 HATEOAS Envelope, Sub-ms RAM Caching, Real-time SSE Stream, and Docker CI/CD.

3. Sprint Execution Lifecycle

Sprint Phase	User Stories & Objectives	Delivered Artifacts	Definition of Done
Sprint 1 ***(Core MVP)***	• Student CRUD endpoints • Relational MongoDB schemas • Basic React table dashboard	• /api/v1/students • Mongoose Student schema • Dashboard Component	Clean build & baseline REST routes operational.
Sprint 2 ***(Security & UI)***	• Google/GitHub OAuth 2.0 • Dual JWT Refresh Tokens • Soft Delete Trash Console • WebGL Shaders & Glassmorphism	• /api/v1/auth/social • Trash Console (/trash) • AuthPage.tsx • Liquid Metal Shader	Zero security flaws & 60fps animations.
Sprint 3 ***(Enterprise & DevOps)***	• API v2 HATEOAS Envelope • Sub-ms RAM Caching Engine • Real-time SSE Stream • Docker & CI/CD Setup	• /api/v2/students • RAM Cache (X-Cache) • Dockerfile & Compose • GitHub Actions workflow	100% Jest tests passing & Docker Compose green.

4. System Architecture & Design Patterns

EduBase implements a **3-Tier Layered Enterprise Architecture** comprising a Presentation Tier (React 18, Tailwind, Framer Motion), Logic Tier (Express 5, RAM Cache, SSE Stream, Dual JWT Auth), and Data Storage Tier (MongoDB Atlas with Relational Mongoose populate joins).

- **MVC Pattern:** Decouples controllers, data models, and view components.
- **Middleware Chain Pattern:** Sequential security processing (Helmet, Rate Limiting, JWT Verification).
- **Observer Pattern (SSE):** HTTP event stream pushing live database mutation updates to client UI.

5. Verification & Validation (V-Model Integration)

Requirement Under Test	Target Test Suite File	Validation Result
API Sanity & Health Check	**tests/api.test.js (L38-L42)**	PASSED (200 OK)
JWT Authorization Protection	**tests/api.test.js (L44-L60)**	PASSED (401 Unauthorized)
Custom 404 Route Handling	**tests/api.test.js (L62-L66)**	PASSED (404 Not Found)
Duplicate Student ID Rejection	**tests/api.test.js (L81-L92)**	PASSED (400 Bad Request)
Age Boundary Validation (16–90)	**tests/api.test.js (L95-L121)**	PASSED (400 Bad Request)
Silent Refresh Token Renewal	**tests/api.test.js (L125-L133)**	PASSED (200 OK)
API v2 HATEOAS Envelope Contract	**tests/api.test.js (L135-L144)**	PASSED (200 OK)

6. DevOps, CI/CD Pipeline & Deployment

• **GitHub Actions CI/CD:** Workflow in .github/workflows/ci-cd.yml provisions a containerized MongoDB service, runs automated Jest API tests, and verifies the frontend production build on every push.

• **Docker Multi-Container Setup:** Configured with Dockerfile and docker-compose.yml orchestrating Backend (Port 5050), Frontend (Port 3000), and MongoDB (Port 27017).

7. Conclusion & Faculty Sign-Off

The **EduBase** project successfully demonstrates a rigorous software engineering process by combining the **Agile Model (Scrum)** for adaptive project execution and **Component-Based Development (CBD)** for scalable technical design, verified through automated integration testing and continuous CI/CD deployment.

Lead System Architect: Yashwanth

Signature: ________________________

UI/UX & Frontend Lead: Gagan Aditya

Signature: ________________________

Faculty Evaluator: Department of CS & Engineering

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